

Mth-202 Important Mid Term
Mcq's Solution 100% Correct :
Solve By Vu-Topper RM!!

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Question No:1 (Marks:1) Vu-Topper RM

Range of function $f(x) = (e^x)$ is _____

➤ **Set of positive real numbers**

Question No:2 (Marks:1) Vu-Topper RM

Composite relation symbolically written as _____

➤ **$SoR = \{(a,c) \in A \times C, (c,b) \in C \times B, (a,b) \in R \text{ and } (b,c) \in S\}$**

Question No:3 (Marks:1) Vu-Topper RM

If $x \equiv 17 \pmod{5}$ which of the following integers are valid solution for x ?

➤ **12**

Question No:4 (Marks:1) Vu-Topper RM

Range of the relation $\{(0,1), (3,22), (90,34)\}$

➤ **$\{1,22,34\}$**

Question No:5 (Marks:1) Vu-Topper RM

Let $A = \{0,1,2\}$ and $R = \{(0,2), (1,1), (2,0)\}$ be a relation on A. The which of the following ordered pairs are needed to make it transitive?

➤ **$(0,0)$ and $(2,2)$**

Question No:6 (Marks:1) Vu-Topper RM

Operation of subtraction is a binary operation on the set of _____

➤ **Integers**

Question No:7 (Marks:1) Vu-Topper RM

Let $S=R$ and define the 'square' relation $R = \{(x,y) | x^2 = y^2\}$. The square relation is an _____ relation

➤ **Equivalence relation**

Question No:8 (Marks:1) Vu-Topper RM

The logic gate NOT is a unary operation on $\{0,1\}$.

➤ **True**

Question No:9 (Marks:1) Vu-Topper RM

Let $A = \{1,2\}$, then $P(A) =$ _____

➤ **$\{\{\}, \{1\}, \{2\}, \{1,2\}\}$**

Question No:10 (Marks:1) Vu-Topper RM

If a relation R is reflexive, anti symmetric and transitive then which of the following is not true for the inverse relation.

➤ **Inverse relation will be irreflexive.**

Question No:11 (Marks:1) Vu-Topper RM

Let R be a binary relation on a set A, R is anti-symmetric iff _____

➤ **$a, b \in A$ if $(a,b) \in R$ and $(b,a) \in R$ then $a=b$**

Question No:12 (Marks:1) Vu-Topper RM

“-“ is a binary operation on the set of integers $\mathbb{Z}$.

➤True

Question No:13 (Marks:1) Vu-Topper RM

The inverse relation R^{-1} from B to A is defined as _____

➤ $R^{-1} = \{(b,a) \in B \times A \mid (a,b) \in R\}$

Question No:14 (Marks:1) Vu-Topper RM

Which of the following is always true for the matrix representation of a symmetric relation?

➤Matrix is equal to its transpose

Question No:15 (Marks:1) Vu-Topper RM

Let $A = \{1,2,3,4\}$ and let R and S be transitive binary relations on A defined as; $R = \{(1,2), (1,3), (2,2), (3,3), (4,2), (4,3)\}$ and $S = \{(2,1), (2,4), (3,3)\}$ then $R \cup S = \{(1,2), (1,3), (2,1), (2,2), (2,4), (3,3), (4,2), (4,3)\}$

➤ $R \cup S$ is transitive

Question No:16 (Marks:1) Vu-Topper RM

Let $S=R$ and define the ‘square’ relation $R = \{(x,y) \mid x^2 = y^2\}$. The square is an _____ relation.

➤Equivalence relation

Question No:17 (Marks:1) Vu-Topper RM

If $x \equiv -10 \pmod{15}$. Which of the following integers are valid solution for x ?

➤5

Question No:18 (Marks:1) Vu-Topper RM

Let R be a binary relation on a set A . If R is anti symmetric then _____

➤Inverse of R is anti symmetric

Question No:19 (Marks:1) Vu-Topper RM

If $A = \{1,2,3\}$ is a set and $R = \{(1,2), (2,2), (2,1)\}$ is a relation on A , R is

➤Symmetric

Question No:20 (Marks:1) Vu-Topper RM

Let $A = \{0,1\}$ and $B = \{1\}$. Let R and S be two binary relations on Cartesian product of A and B such that $R = \{(0,1)\}$ and $S = \{(1,1)\}$. Then $R \cap S =$ _____

➤Empty

Question No:21 (Marks:1) Vu-Topper RM

A relation R is said to be _____ iff it is reflexive, antisymmetric and transitive.

➤Partial order Relation

Question No:22 (Marks:1) Vu-Topper RM

Let $X=\{1,2,3\}$ and $Y=\{7,8,9\}$ and let f be function defined from X to Y such that f is onto then which of the following statement about f is true?

➤ **Co-domain of f must contain 1 element**

Question No:23 (Marks:1)

Vu-Topper RM

The function fog and gof are always equal

➤ **False**

Question No:24 (Marks:1)

Vu-Topper RM

If a relation $R=\{(1,2),(2,3),(3,4),(4,1),(2,2)\}$ is given then which of the following is true about this relation.

➤ **R is reflexive**

Question No:25 (Marks:1)

Vu-Topper RM

A set is called countable if , and only if, it is _____

➤ **finite**

Question No:26 (Marks:1)

Vu-Topper RM

Let $f(x) = x^2 - 1$ define function f from R to R and $c=2$ be any scalar, then $c \cdot f(x)$ is _____

➤ **$2x^2 + 2$**

Question No:27 (Marks:1)

Vu-Topper RM

The set Z of all integers is _____

➤ **Countable**

Question No:28 (Marks:1)

Vu-Topper RM

Let R be a binary relation on a set A . If R is anti symmetric then _____

➤ **Inverse of R is symmetric**

Question No:29 (Marks:1)

Vu-Topper RM

For $(2x-3, 4y+2) = (1,10)$. What will be the value of x and y ?

➤ **$(2,2)$**

Question No:30 (Marks:1)

Vu-Topper RM

Let f and g be the two functions from R to R defined by $f(x) = |x|$ and $g(x) = \text{square root of } x^2$ for all $x \in R$. Then _____

➤ **$f(x)$ is not equal to $g(x)$**

Question No:31 (Marks:1)

Vu-Topper RM

If a set A has 15 elements then $P(A)$ (power set of A) has _____ elements.

➤ **2^{15}**

Question No:32 (Marks:1)

Vu-Topper RM

For the relation below to be a function, x cannot be what values $\{(12,14),(13,5),(-2,7),(x,13)\}$?

➤ **x cannot be 12, 13, or -2**

Question No:33 (Marks:1)

Vu-Topper RM

Let the set $A = \{1,2,3,4\}$. Then the relation $\{(2,4),(4,2)\}$ is _____

➤ **Symmetric**

Question No:34 (Marks:1) Vu-Topper RM

For the following relation to be a function, x can not be what values? $R = \{(2,4),(x,1),(4,2),(5,6)\}$.

➤ **x cannot be 2,4 and 5**

Question No:35 (Marks:1) Vu-Topper RM

Vertical line test is used to determine that whether the graph of a relation is a function or not.

➤ **True**

Question No:36 (Marks:1) Vu-Topper RM

The properties of being symmetric and being anti symmetric are _____

➤ **Not negative of each other**

Question No:37 (Marks:1) Vu-Topper RM

The number of elements in $A \times B$ are _____ if A is a set with '5' elements and B is a set with '4' elements.

➤ **20**

Question No:38 (Marks:1) Vu-Topper RM

$R = \{(a,1)(b,2)(c,3)(d,4)\}$ then the inverse of this relation is _____

➤ **$\{(1,a)(2,b)(3,c)(4,d)\}$**

Question No:39 (Marks:1) Vu-Topper RM

Logic gate NOT does not define a binary operation on $(0,1)$ because _____

➤ **It takes a single input and gives a single output**

Question No:40 (Marks:1) Vu-Topper RM

How many real numbers exist between 1 and 5

➤ **3**

Question No:41 (Marks:1) Vu-Topper RM

The number π is

➤ **Irrational**

Question No:42 (Marks:1) Vu-Topper RM

The number square root 2 is

➤ **Irrational**

Question No:43 (Marks:1) Vu-Topper RM

Range the relation $\{(0,1)(3,22),(90,34)\}$

➤ **$\{0,3,90\}$**

Question No:44 (Marks:1) Vu-Topper RM

Supherical coordinate 0 is related to the cylindrical coordinate as _____

➤@=@

Question No:45 (Marks:1) Vu-Topper RM

Operation of subtraction is a binary operation on the set_____

➤Integers

Question No:46 (Marks:1) Vu-Topper RM

Let $A = \{1,2,3,4\}$ and $R = \{(1,2),(2,3),(3,3),(3,4)\}$ be a relation on A . Then which one of the following ordered pair has made R not an irreflexive relation?

➤ $(3,3)$

Question No:47 (Marks:1) Vu-Topper RM

Input values of the function are called the _____

➤Domain

Question No:48 (Marks:1) Vu-Topper RM

Range of function $f(x) = |x|$ will be

➤Set of positive real numbers

Question No:49 (Marks:1) Vu-Topper RM

Which of the following is not a binary operation on the set of integers?

➤Division

Question No:50 (Marks:1) Vu-Topper RM

In the matrix representation of an irreflexive relation all the entries in the main diagonal are

➤0

Question No:51 (Marks:1) Vu-Topper RM

If the partition set of A is $\{A_1, A_2\}$ then

➤ $A_1 \cap A_2 = \text{not empty set}$

Question No:52 (Marks:1) Vu-Topper RM

Let $A = \{a, b\}$ then $P(A) =$

➤{Non empty set, $\{a\}, \{b\}, \{a, b\}$ }

Question No:53 (Marks:1) Vu-Topper RM

Which relations below are not functions?

➤ $\{(13,14), (13,5), (16,7), (18,13)\}$

Question No:54 (Marks:1) Vu-Topper RM

In the directed graph of an antisymmetric relation there is _____ pair of arrows between two distinct elements of the set.

➤No

Question No:55 (Marks:1) Vu-Topper RM

If a relation $R = (1,1), (2,1), (2,2)$ is given then which of the following is not true about this relation

➤ **R is irreflexive**

Question No:56 (Marks:1) Vu-Topper RM

Let R and S be transitive relations on a set A then _____

➤ **Neither R union S is transitive nor R intersection S is transitive**

Question No:57 (Marks:1) Vu-Topper RM

Let $R = \{(1,2), (3,4), (5,6), (7,8)\}$. Domain of the inverse of the relation is _____

➤ **{2,4,6,8}**

Question No:58 (Marks:1) Vu-Topper RM

Let $A = \{1,2,3,4,5\}$ and $B = \{4,9,16,17,25\}$. Then the relation $R = \{(2,4), (3,9), (4,16), (9,17)\}$ The inverse of R is'

➤ **{(4,2), (9,3), (16,4), (17,3)}**

Question No:59 (Marks:1) Vu-Topper RM

Let R be a relation on a set A. If R is symmetric then its compliment is

➤ **Irreflexive**

Question No:60 (Marks:1) Vu-Topper RM

Which is not a binary operation on the set of natural numbers N ?

➤ **Subtraction**

Question No:61 (Marks:1) Vu-Topper RM

If a relation $R = \{(1,1), (2,1), (1,2), (2,2)\}$ is given then which of the following is not true about this relation.

➤ **R is irreflexive**

Question No:62 (Marks:1) Vu-Topper RM

$R = \{(a,1), (b,2), (c,3), (d,4)\}$ then the inverse of this relation is _____

➤ **{(1,a), (2,b), (3,c), (4,d)}**

Question No:63 (Marks:1) Vu-Topper RM

For any set A, the Cartesian product of A and A is known as _____

➤ **Universal relation**

Question No:64 (Marks:1) Vu-Topper RM

Let $A = \{p, q, r, s\}$ and define a relation R on A by $R = \{(p,p), (p,r), (q,r), (q,s), (r,s)\}$ Then which one of the following is the correct statement about R:

➤ **R is not reflexive**

Question No:65 (Marks:1) Vu-Topper RM

$A = \{1,2\}$ $B = \{3,4\}$, $R = \{(1,3), (2,4)\}$. Then the complement of R is _____.

➤ **{(1,4), (2,3)}**

Question No:66 (Marks:1) Vu-Topper RM

Domain of a relation symbolically written as_____.

➤ $\text{Dom}(R)=\{a \in A \mid (a,b) \in R\}$

Question No:67 (Marks:1) Vu-Topper RM

Let $X=\{2,4,5\}$ and $Y=\{1,2,4\}$ and R be a relation from X to Y defined by $R=\{(2,4)(4,1)(a,2)\}$. For what value of 'a' the relation R is a function?

➤5

Question No:68 (Marks:1) Vu-Topper RM

Let $A=\{1,2,3,4,5,6,7,8,9\}$, then which of the following sets represent the partition of the set A ?

➤ $A=\{1,3,5,7,9\}$, $B=\{2,4,6\}$, $C=\{8\}$

Question No:69 (Marks:1) Vu-Topper RM

Let $A=\{1,2,3\}$ and $B=\{2,4\}$ then number of binary relations from A to B are _____.

➤64

Question No:70 (Marks:1) Vu-Topper RM

A relation R is said to be _____ iff it is reflexive, antisymmetric and transitive.

➤Partial order Relation

Question No:71 (Marks:1) Vu-Topper RM

Let f be a function from $X=\{2,4,5\}$ to $Y=\{1,2,4,6\}$ defined as: $f=\{(2,6),(4,2),(5,1)\}$. The range of f is _____

➤ $\{1,2,6\}$

Question No:72 (Marks:1) Vu-Topper RM

Let $A=\{0,1,2\}$ and $R=\{(0,2),(1,1),(2,0)\}$ be a relation on A . Then which of the following statement about R is true?

➤ R is symmetric

Question No:73 (Marks:1) Vu-Topper RM

Let $A=\{2,3,4\}$ and $B=\{2,6,8\}$ and let R be the "divides" relation from A to B i.e for all (a,b) belong to (Cartesian product of A and B), aRb iff $a \mid b$ (a divides b). Then

➤ $R=\{(2,2),(2,6),(2,8),(3,6),(4,8)\}$

Question No:74 (Marks:1) Vu-Topper RM

Let $A=\{1,2,3,\dots,50\}$ and $B=\{2,4,6,8,10\}$. Then the Cartesian product of A and B has _____ elements.

➤250

Question No:75 (Marks:1) Vu-Topper RM

In the matrix representation of an reflexive relation all the entries in the main diagonal are _____

➤1

Question No:76 (Marks:1) Vu-Topper RM

Which of the following is not a type of a relation?

➤ **Permutation**

Question No:77 (Marks:1) Vu-Topper RM

Let $X=\{2,4,5\}$ and $Y=\{1,2,4\}$ and R be a relation from X to Y defined by $R=\{(2,4),(4,1),(a,2)\}$. For what value of 'a' the relation R is a function ?

➤ **5**

Question No:78 (Marks:1) Vu-Topper RM

Which of the following is not a representation of a relation?

➤ **Venn diagram**

Question No:79 (Marks:1) Vu-Topper RM

Let $A=\{1,2,3,4\}$ and define the following relations on A . Then

$R=\{(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)\}$ is _____

➤ **R is irreflexive**

Question No:80 (Marks:1) Vu-Topper RM

The range of $f:X \rightarrow Y$ is also called the image of

➤ **True**

Question No:81 (Marks:1) Vu-Topper RM

Which of the following logical connective is not a binary operation ?

➤ **Implication**

Question No:82 (Marks:1) Vu-Topper RM

A set may be divided up into its disjoint subsets, such division is called _____

➤ **Partition**

Question No:83 (Marks:1) Vu-Topper RM

If $A=\{1,2,3\}$ & $B=\{4,5,6\}$ and $R=\{(1,4)(2,5)(3,6)(3,4)\}$ The complementary relation is _____

➤ **$A \times B$ (difference or $-$) R**

Question No:84 (Marks:1) Vu-Topper RM

In matrix representation of a _____ relation, the diagonal entries are always 1.

➤ **Reflexive**

Question No:85 (Marks:1) Vu-Topper RM

R is not symmetric iff there are elements a and b in A such that _____

➤ **(a,b) belongs to R but (b,a) does not belong to R**

Question No:86 (Marks:1) Vu-Topper RM

Which relations below are functions?

$R_1=\{(3,4),(4,5),(6,7),(8,9)\}$

$R_2=\{(3,4),(4,5),(6,7),(3,9)\}$

$R_3=\{(-3,4),(4,-5),(0,0),(8,9)\}$

$R_4=\{(8,11),(34,5),(6,17),(8,19)\}$

➤ **R_1 and R_3 are functions**

Question No:87 (Marks:1) Vu-Topper RM

The logic gate OR and AND are uniary operation on $\{0,1\}$

➤False

Question No:88 (Marks:1) Vu-Topper RM

There is atleast one loop in the graph of an irreflexive relation

➤False

Question No:89 (Marks:1) Vu-Topper RM

There is atleast one loop in the graph of an reflexive relation

➤True

Question No:90 (Marks:1) Vu-Topper RM

A contains 3 elements and B contains 2 elements, then number of subsets of $A*B$ are

➤64

Question No:91 (Marks:1) Vu-Topper RM

Let $A=\{1,2,3\}$ and $B=\{0,1,2\}$ and $C=\{a,b\}$ $R=\{(1,0),(1,2),(3,1),(3,2)\}$ $S=\{(0,b),(1,a),(2,b)\}$
composite of R and S=_____

➤ $\{(1,b),(1,a),(3,a),(3,b)\}$

Question No:92 (Marks:1) Vu-Topper RM

If R is transitive then the inverse relation will be transitive

True

Question No:93 (Marks:1) Vu-Topper RM

The number of elements in the power set of P(not empty set) denoted by $P(P(\text{not empty set}))$ is

➤2

Question No:94 (Marks:1) Vu-Topper RM

The function defined from Z to Z as $f(x)=1/(x+2)(x-2)$ is not well defined because _____

➤Function is not defined at $x=2$ and $x=-2$

Question No:95 (Marks:1) Vu-Topper RM

Rang of relation $\{(0,1),(3,22),(90,34)\}$ is _____

➤{1, 22, 34}

Question No:96 (Marks:1) Vu-Topper RM

The number of elements in the power set of P(not empty set) denoted by $P(\text{not empty set})$ is

➤1

Question No:97 (Marks:1) Vu-Topper RM

Let $R=\{(1,2),(3,4),(5,6),(7,8)\}$. Domain of inverse of the relation is _____

➤{2,4,6,8}

Question No:98 (Marks:1) Vu-Topper RM

The relation 'divides' on the set of integers is _____

➤ **A symmetric relation**

Question No:99 (Marks:1) Vu-Topper RM

Operation of subtraction is a binary operation on the set of _____

➤ **Integers**

Question No:100 (Marks:1) Vu-Topper RM

If R is transitive then the inverse relation will be transitive.

➤ **True**

Question No:101 (Marks:1) Vu-Topper RM

Let $A = \{1, 2, 3, 4\}$ and define the relation R on A by $R = \{(1, 2), (2, 3), (3, 3), (3, 4)\}$.

Then _____

➤ **R is both reflexive and irreflexive**

Question No:102 (Marks:1) Vu-Topper RM

A set may be divided up into its disjoint subsets, such division is called

➤ **Partition**

Question No:103 (Marks:1) Vu-Topper RM

If a set A contains n elements then the number of elements in its power set $P(A)$ is _____

➤ **2^n**

Question No:104 (Marks:1) Vu-Topper RM

Range of a relation symbolically written as _____.

➤ **$\text{Ran } R = \{b \in B \mid (a, b) \in R\}$**

Question No:105 (Marks:1) Vu-Topper RM

Let R be a binary relation on a set A. If R is anti symmetric then _____.

➤ **Inverse of R is anti symmetric**

Question No:106 (Marks:1) Vu-Topper RM

Let A be a set with m elements and B be a set with n elements then the number of elements in $A \times B$ are _____

➤ **m.n**

Question No:107 (Marks:1) Vu-Topper RM

Let $A = \{1, 2, 3, 4\}$ and define the following relation on A. Then $R = \{(1, 3), (2, 2), (2, 4), (3, 1), (4, 2)\}$
Is _____

➤ **R is symmetric**

Question No:108 (Marks:1) Vu-Topper RM

If $A = \{1, 2, 3\}$ & $B = \{4, 5, 6\}$ and $R = \{1, 4\}, \{2, 5\}, \{3, 6\}, \{3, 4\}$. The complementary relation is

➤ **$A \times B - R$**

Question No:109 (Marks:1) Vu-Topper RM

Let R be a relation on a set A . If R is reflexive then its complement is _____

➤ **Irreflexive**

Question No:110 (Marks:1)

Vu-Topper RM

$25 \equiv 1 \pmod{3}$ means that 3 divides _____

➤ **25-1**

Question No:111 (Marks:1)

Vu-Topper RM

Let R be the universal relation on a set A then which one of the following statement about R is true ?

➤ **R is reflexive, symmetric and transitive**

Question No:112 (Marks:1)

Vu-Topper RM

Domain of a relation symbolically written as _____

➤ **$\text{Dom}(R) = \{a \in R \mid (a,b) \in R\}$**

Question No:113 (Marks:1)

Vu-Topper RM

Let R be a relation on a set A . R is transitive if and only if for all $a,b,c \in A$ then

➤ **$(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$**

Question No:114 (Marks:1)

Vu-Topper RM

Let A be a non-empty set and $P(A)$ the power set of A . Define the 'subset' relation, $\subseteq$ as follows for all $X, Y \in P(A)$, $X \subseteq Y \iff$ for all x , if $x \in X$ then $x \in Y$. Then $\subseteq$ is _____

➤ **$\subseteq$ is partial order relation**

Question No:115 (Marks:1)

Vu-Topper RM

Define a relation $R = \{(1,1), (2,2), (3,3), (1,3)\}$ the relation is

➤ **R is reflexive and transitive**

Question No:116 (Marks:1)

Vu-Topper RM

Let R be a relation on a set A . R is transitive if and only if for all $a,b,c \in A$ then.

➤ **$(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$**

Question No:117 (Marks:1)

Vu-Topper RM

Let R and S be reflexive relations on a set A then $R \cap S$ is reflexive

➤ **True**

Question No:118 (Marks:1)

Vu-Topper RM

A function whose range consists of only one element is called _____

➤ **One to one function**

Question No:119 (Marks:1)

Vu-Topper RM

The set $\mathbb{Z}$ of all integers is _____

➤ **Countable**

Question No:120 (Marks:1)

Vu-Topper RM

Let $X = \{1, 5, 9\}$ and $Y = \{3, 4, 7\}$, Define a function f from X to Y such that $f(1) = 7, f(5) = 3, f(9) = \underline{\hspace{2cm}}$. Which is true $f(9)$ to make it a one-to-one

(injective) function?

➤4

Question No:121 (Marks:1) Vu-Topper RM

What will be the fourth term of the following sequence $1/2, 2/3, 3/4$ _____?

➤4/5

Question No:122 (Marks:1) Vu-Topper RM

The value of $6!$ =

➤720

Question No:123 (Marks:1) Vu-Topper RM

A constant function is one to one iff its _____ is a singleton.

➤Domain

Question No:124 (Marks:1) Vu-Topper RM

A constant function is onto iff its _____ is a singleton.

➤Co-domain

Question No:125 (Marks:1) Vu-Topper RM

Number of one to one functions from $X = \{a, b\}$ to $Y = \{u, v\}$ are equal to _____.

➤2

Question No:126 (Marks:1) Vu-Topper RM

If f is defined recursively by $f(0) = -1$ and $f(n+1) = f(n) + 3$, then $f(2) =$ _____.

➤5

Question No:127 (Marks:1) Vu-Topper RM

A function whose inverse function exists is called a/an _____.

➤Invertible function

Question No:128 (Marks:1) Vu-Topper RM

Let $f(2) = 3$, $g(2) = 3$, $f(4) = 1$ and $g(4) = 2$ then the value of $f \circ g(4)$ is.....

➤3

Question No:129 (Marks:1) Vu-Topper RM

Let $A = \{1, 2, 3, 4\}$ and $B = \{7\}$ then the constant function from A to B is _____.

➤Onto

Question No:130 (Marks:1) Vu-Topper RM

Composition of a function is a commutative operation.

➤False

Question No:131 (Marks:1) Vu-Topper RM

Composition of a function is not a commutative operation

➤True

Question No:132 (Marks:1) Vu-Topper RM

The sum of first five whole number is _____.

➤10

Question No:133 (Marks:1) Vu-Topper RM

If f and g are two one-to-one functions, then their composition that is $g \circ f$ is one-to-one.

➤True

Question No:134 (Marks:1) Vu-Topper RM

Inverse of a surjective function is always a function.

➤False

Question No:135 (Marks:1) Vu-Topper RM

Inverse of a surjective function may not be a function

➤True

Question No:136 (Marks:1) Vu-Topper RM

Let $X = \{1, 2, 3, 4\}$ and $Y = \{7, 8, 9\}$ and let f be function defined from X to Y such that f is onto then which of the following statement about f is true?

➤Co-domain of f must contain 3 elements

Question No:137 (Marks:1) Vu-Topper RM

If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are both onto functions. Then $g \circ f: X \rightarrow Z$ is _____

➤onto

Question No:138 (Marks:1) Vu-Topper RM

If f and g are two one-to-one functions, then their composition $g \circ f$ is _____

➤One-to-one

Question No:139 (Marks:1) Vu-Topper RM

If $f: W \rightarrow X$, $g: X \rightarrow Y$, and $h: Y \rightarrow Z$ are functions, then _____

➤ $(h \circ g) \circ f = h \circ (g \circ f)$

Question No:140 (Marks:1) Vu-Topper RM

Cardinality of positive prime numbers less than 20 is _____.

➤8

Question No:141 (Marks:1) Vu-Topper RM

If $f(x) = \sin^{-1}(x)$ and $g(x) = \sin x$ then $g \circ f(x)$ is _____.

➤X

Question No:142 (Marks:1) Vu-Topper RM

$0! =$ _____.

➤1

Question No:143 (Marks:1) Vu-Topper RM

An important data type in computer programming consists of _____.

➤Finite sequences

Question No:144 (Marks:1) Vu-Topper RM

Let $f(x)=2x$ and $g(x)=x+2$ define functions f and g from R to R , then $(f-g)(x)$ is _____.

➤x-2

Question No:145 (Marks:1) Vu-Topper RM

The total number of terms in an arithmetic series $0+5+10+15+....50$ are _____.

➤11

Question No:146 (Marks:1) Vu-Topper RM

$9!/6!=$ _____

➤504

Question No:147 (Marks:1) Vu-Topper RM

Let $f(x)=3x$ and $g(x)=3x-2$ define functions f and g from R to R , THEN $(F+G)(X)$ is _____

➤6x-2

Question No:148 (Marks:1) Vu-Topper RM

If f is a bijective function then $(f^{-1}f(x))$ is equal to

➤X

Question No:149 (Marks:1) Vu-Topper RM

A sequence whose terms alternate in sign is called an _____

➤Alternating sequence

Question No:150 (Marks:1) Vu-Topper RM

Common ratio in the sequence “4, 16, 64, 256,...” is.....12.

➤4

Question No:151 (Marks:1) Vu-Topper RM

0.8181818181 is a infinite geometric series.

➤True

Question No:152 (Marks:1) Vu-Topper RM

The word ‘algorithm’ refers to a step-by-step method for performing some action.

➤True

Question No:153 (Marks:1) Vu-Topper RM

A predicate become _____ when its variables are given specific values

➤Sentence

Question No:154 (Marks:1) Vu-Topper RM

The sum of two irrational numbers must be an irrational number.

➤False

Question No:155 (Marks:1) Vu-Topper RM

The sum of two irrational numbers need not be irrational number

➤True

Question No:156 (Marks:1) Vu-Topper RM

The division by zero is allowed in mathematics.

➤False

Question No:157 (Marks:1) Vu-Topper RM

The product of any two consecutive positive integers is divisible by 2

➤True

Question No:158 (Marks:1) Vu-Topper RM

If 'n' is an odd integer then n^3+n is _____.

➤Even

Question No:159 (Marks:1) Vu-Topper RM

For integers a,b,c, If divides b and a divides c, then a divides (a+b).

➤False

Question No:160 (Marks:1) Vu-Topper RM

Quotient remainder theorem states that for any positive integer d, there exist unique integer q and r such that _____ and $0 \leq r < d$

➤ $N=d.q+r$

Question No:161 (Marks:1) Vu-Topper RM

A rule that assigns a numerical value to each outcome in a sample space is called

➤Random variable

Question No:162 (Marks:1) Vu-Topper RM

If A and B are two disjoint (mutually exclusive) events then $P(A \cup B) =$

➤ $P(A) + P(B)$

Question No:163 (Marks:1) Vu-Topper RM

How many ways are there to select five players from a 10 member tennis team to make a trip to a match to another school?

➤ $C(10,5)$

Question No:164 (Marks:1) Vu-Topper RM

The expectation of x is equal to

➤ $\sum xf(x)$

Question No:165 (Marks:1) Vu-Topper RM

If $P(A \cap B) = P(A)P(B)$ THEN THE events A and B are called

➤Independent

Question No:166 (Marks:1) Vu-Topper RM

A walk that starts and ends at the same vertex is called.

➤None optins

Question No:167 (Marks:1) Vu-Topper RM

How many integers from 1 through 1000 are neither multiple of 3 nor multiple of 5

➤497

Question No:168 (Marks:1) Vu-Topper RM

What is the probability of getting a number greater than 4 when a die is thrown?

➤3/5

Question No:169 (Marks:1) Vu-Topper RM

Eater formula for graphs is_____.

➤ $F=e-v+2$

Question No:170 (Marks:1) Vu-Topper RM

$X+a, x+3a, x+5a, \dots$ is an _____

➤Arithmetic sequence

Question No:171 (Marks:1) Vu-Topper RM

Composition of a function is a commutative operation.

➤False

Question No:172 (Marks:1) Vu-Topper RM

Real valued function is a function that assigns _____ to each member of its domain.

➤Only a real number

Question No:173 (Marks:1) Vu-Topper RM

Let $X=\{1,2,3,4\}$ and a function 'f' defined on X $f(1)=1, f(2)=2, f(3)=3, f(4)=4$ then _____

➤F is an identity function

Question No:174 (Marks:1) Vu-Topper RM

A constant function is surjective if and only if _____

➤The co-domain consists of a single element

Question No:175 (Marks:1) Vu-Topper RM

Cardinality means the total number of elements in a set.

➤True

Question No:176 (Marks:1) Vu-Topper RM

If $f(x)=2x$ and $g(x)=x$ then $g(f(x))$ is _____.

➤ $2x^2$

Question No:177 (Marks:1) Vu-Topper RM

Let $f:R \rightarrow R$ is one to one function then c,f, c is not equal 0 is also one to one function.

➤True

Question No:178 (Marks:1) Vu-Topper RM

Let $X=\{1,2,3,4\}$ and a function 'f' defined from X to X by $f(1)=1, f(2)=1, f(3)=1, f(4)=1$ then which of the following is true?

➤F is a constant function

Question No:179 (Marks:1) Vu-Topper RM

If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

➤True

Question No:180 (Marks:1) Vu-Topper RM

Which of the following is not correct for a 'sequence'?

➤A sequence is a relation whose domain is the set of natural numbers

Question No:181 (Marks:1) Vu-Topper RM

$F(x)=x^2$ is not one to one function from R to R^+

➤True

Question No:182 (Marks:1) Vu-Topper RM

Let $f: R \rightarrow R$ is one to one function then $c, f \circ c$ is not equal to c is also one to one function.

➤True

Question No:183 (Marks:1) Vu-Topper RM

Let $f(x)=x+2$ then $f^{-1}(x)$ is _____

➤x-2

Question No:184 (Marks:1) Vu-Topper RM

Let $f(x)=x^2+1$ define functions f from R to R and $c=2$ be any scalar, then $c, f(x)$ is _____.

➤ $2x^2-1$

Question No:185 (Marks:1) Vu-Topper RM

A function $F: R \rightarrow R$ defined by $f(x) = \text{square root } x$ is a real valued function.

➤False

Question No:186 (Marks:1) Vu-Topper RM

If $g: R \rightarrow R$ defined by $g(x)=e^x$ is a real valued function of a real variable.

➤True

Question No:187 (Marks:1) Vu-Topper RM

A function $F: R \rightarrow R$ defined by $f(x) = \log x$ is a real valued function

➤True

Question No:188 (Marks:1) Vu-Topper RM

$\frac{1}{2}$, then 3rd term of sequence is _____.

➤ $\frac{1}{2}$

Question No:189 (Marks:1) Vu-Topper RM

The process of defining an object in terms of smaller versions of itself is called recursion.

➤True

Question No:190 (Marks:1) Vu-Topper RM

Which of the following is not correct for a 'sequence'?

➤A sequence is a relation whose domain is the set of natural numbers

Question No:191 (Marks:1) Vu-Topper RM

A set is called countable if, and only if, it is _____.

➤Finite and countable infinite.....both

Question No:192 (Marks:1) Vu-Topper RM

A set that is not countable is called _____.

➤Uncountable

Question No:193 (Marks:1) Vu-Topper RM

A sequence whose terms alternate in sign is called an _____.

➤Alternating sequence

Question No:194 (Marks:1) Vu-Topper RM

Let $f: R \rightarrow R$ is one to one function then c, f is also one to one function for _____.

➤C is not equal 0

Question No:195 (Marks:1) Vu-Topper RM

Let $f(x) = x+3$ then $f^{-1}(x)$ is _____.

➤x-3

Question No:196 (Marks:1) Vu-Topper RM

Let f and g be two functions defined by $f(x) = x+2$ and $g(x) = 2x+1$. Then the composition of f and g is _____.

➤ $2x+3$

Question No:197 (Marks:1) Vu-Topper RM

Number of one to one functions from $X = \{a, b\}$ to $Y = \{u, v\}$ are equal to _____.

➤2

Question No:198 (Marks:1) Vu-Topper RM

The fibonacci sequence is defined as $F_0=1, F_1=1, F_k=F_{k-1}+F_{k-2}$ for all integers $k \geq 2$ then which of the following is true for F_2

➤ $F_2-F_1 = 2+1=3$

Question No:199 (Marks:1) Vu-Topper RM

$x+a, x+3a, x+5a, \dots$ is a/an _____.

➤Arithmetic sequence

Question No:200 (Marks:1) Vu-Topper RM

Inverse of a function may not be a function.

➤True

Question No:201 (Marks:1) Vu-Topper RM

In the following sequence $a_k = K/(k+1)$, for $k=1$, a_1 will be _____.

➤ $\frac{1}{2}$

Question No:202 (Marks:1) Vu-Topper RM

If $f(x)=x$ and $g(x)=-x$ are both one to one function then $(f+g)(x)$ is also one to one function.

➤ False

Question No:203 (Marks:1) Vu-Topper RM

If $f(x)=x$ and $g(x)=-x$ are both one to one function then $(f+g)(x)$ is not one to one function.

➤ True

Question No:204 (Marks:1) Vu-Topper RM

The function 'f' and 'g' are inverse of each other if and only if their composition gives _____.

➤ Identity function

Question No:205 (Marks:1) Vu-Topper RM

Which of the following set is the domain of a sequence?

➤ Set of real numbers

Question No:206 (Marks:1) Vu-Topper RM

Let C is defined as the set of all countries in the world then C is a _____.

➤ Finite set

Question No:207 (Marks:1) Vu-Topper RM

A constant function is surjective if and only if _____.

➤ The co domain consists of a single element

Question No:208 (Marks:1) Vu-Topper RM

Let $X=\{1,2,3,4\}$ and a function 'f' defined from X to X by $f(1)=1$, $f(2)=1$, $f(3)=1$, $f(4)=1$ then which of the following is true?

➤ F is a constant function

Question No:209 (Marks:1) Vu-Topper RM

The composition of function is always

➤ Associative

Question No:210 (Marks:1) Vu-Topper RM

A set is countably infinite if, and only if, it has the same cardinality as the set of

➤ Positive integers

Question No:211 (Marks:1) Vu-Topper RM

Two functions 'f' and 'g' from 'X' to 'Y' are said to be equal if and only if _____.

➤ $f(x)=g(x)$ for all 'x' belongs to X

Question No:212 (Marks:1) Vu-Topper RM

$Y=x^3$ is a graph of bijective function from R to R.

➤True

Question No:213 (Marks:1) Vu-Topper RM

Domain and range are same for_____.

➤Identity function

Question No:214 (Marks:1) Vu-Topper RM

Let $X = \{1, 2, 3, 4\}$ and a function 'f' defined on X by $f(1)=1, f(2)=2, f(3)=3$

➤F is an identity function

Question No:215 (Marks:1) Vu-Topper RM

Composition of a function is a commutative operation.

➤True

Question No:216 (Marks:1) Vu-Topper RM

Inverse of a surjective function is always a function.

➤False

Question No:217 (Marks:1) Vu-Topper RM

A function whose inverse function exists is called a/an_____.

➤Invertible

Question No:218 (Marks:1) Vu-Topper RM

Given a set X define a function I from X to X by $i(x)=x$ from all x belonging to X .

Then_____.

➤I is both injective and surjective

Question No:219 (Marks:1) Vu-Topper RM

Let $f: R \rightarrow R$ is a one to one function then c, f is also one to one function for.

➤C is not equal 0

Question No:220 (Marks:1) Vu-Topper RM

Let $X = \{1, 5, 9\}$ and $Y = \{3, 4, 7\}$. Define a function f from X to Y such that $f(1)=7, f(5)=3, f(9)=$ _____. Which is true for f(9) to make it a one-to-one (injective) function?

➤4

Question No:221 (Marks:1) Vu-Topper RM

Which of the following is not a predecessors of a_k ?

➤ A_{k+1}

Question No:222 (Marks:1) Vu-Topper RM

Two functions 'f' and 'g' from X to Y are said to be equal if and only if_____.

➤ $F(x)=g(x)$ for all 'x' belongs to x

Question No:223 (Marks:1) Vu-Topper RM

The two functions 'f' and 'g' are equal if_____.

➤ $F(x) = 3x$ and $g(x) = \frac{6x^2 + 3x}{2x^2 + 1}$ for all $x \in R$

Question No:224 (Marks:1) Vu-Topper RM

If first term of a geometric sequence is 2 and common ratio is $\frac{1}{2}$, then 3rd ter, of sequence.

➤ $\frac{1}{4}$

Question No:225 (Marks:1) Vu-Topper RM

$Y = \sqrt{x}$ is an _____ function form R^+ to R

➤ **One to one function**

Question No:226 (Marks:1) Vu-Topper RM

$Y = x^2$ is an _____ function form R to R^+

➤ **NOT ONE TO ONE FUNTION**

Question No:227 (Marks:1) Vu-Topper RM

If a function $(g \circ f)(x) : X \rightarrow Z$ is defined as $(g \circ f)(x) = g(f(x))$ for all $x \in X$, Then the function _____.

➤ **(gof)**

Question No:228 (Marks:1) Vu-Topper RM

If 0 is the first term and -2 be the common difference of an arithmetic series, then the sum of first five terms of series is _____.

➤ **-20**

Question No:229 (Marks:1) Vu-Topper RM

If $f(x) = \sin^{-1}(x)$ and $g(x) = \sin x$ then $g \circ f(x)$ is _____.

➤ **X**

Question No:230 (Marks:1) Vu-Topper RM

$f : X \rightarrow Y$ that is both one to one and onto is called a _____.

➤ **Bijective function**

Question No:231 (Marks:1) Vu-Topper RM

What does 'y' denotes in a geometric sequence?

➤ **Common ratio**

Question No:232 (Marks:1) Vu-Topper RM

Let g be a function defined by $g(x) = x + 1$. Then the composition of $(g \circ g)$.

➤ **X+2**

Question No:233 (Marks:1) Vu-Topper RM

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

➤ **True**

Question No:234 (Marks:1) Vu-Topper RM

Which of the following is true for the following sequence?

➤ **If n is even, then $C_n = 2$ and if n is odd, then $C_n = 0$**

Question No:232 (Marks:1) Vu-Topper RM

The function 'f' and 'g' are inverse of each other if and only if their composition gives_____.

➤Identity function

Question No:233 (Marks:1) Vu-Topper RM

N! is defined to be _____.

➤The product of the integers from 1 to n

Question No:234 (Marks:1) Vu-Topper RM

Let $A = \{1, 2, 3, 4\}$ and $B = \{7\}$ then the constant function from A to B

➤Both one to one and onto

Question No:235 (Marks:1) Vu-Topper RM

A set is called countable if, and only if, it is _____.

➤Countably infinite and finite

Question No:236 (Marks:1) Vu-Topper RM

If $f(x) = x$ and $g(x) = -x$ are both one to one functions then $(f+g)(x)$ is also one to one function.

➤False

Question No:237 (Marks:1) Vu-Topper RM

If a is the 1st term and d be the common difference of an arithmetic sequence then the sequence is a, a+d, a+2d, a+3d....

➤True

Question No:238 (Marks:1) Vu-Topper RM

$x+a, x+3a, x+5, \dots$ is a/an _____.

➤Arithmetic sequence

Question No:239 (Marks:1) Vu-Topper RM

inverse of an injective function may not be a function.

➤True

Question No:240 (Marks:1) Vu-Topper RM

$y=x^3$ is a graph of bijective function from R to R

➤False

Question No:241 (Marks:1) Vu-Topper RM

Two functions 'f' and 'g' from x to y are said to be equal if and only if _____.

➤ $F(x)=g(x)$ for all 'x' belongs to x

Question No:242 (Marks:1) Vu-Topper RM

Common ratio in sequence '36, 12, 4, 4/3,' is

➤1/3

Question No:243 (Marks:1) Vu-Topper RM

A set is called finite if, and only if, is the _____ or there is _____.

➤ **Empty set or one-to-one**

Question No:244 (Marks:1) Vu-Topper RM

Let $f(x)=x^2-1$ and $g(x)=x+1$ define functions f and g from $\mathbb{R}$ to $\mathbb{R}$, then $(f/g)x$

➤ **$x-1$**

Question No:245 (Marks:1) Vu-Topper RM

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

➤ **True**

Question No:246 (Marks:1) Vu-Topper RM

Let g be a function defined by $g(x) = x+1$. Then the composition of (gog) .

➤ **$x+2$**

Question No:247 (Marks:1) Vu-Topper RM

$F:X \rightarrow Y$ that is both one to one and onto is called a _____.

➤ **Bijjective function**

Question No:248 (Marks:1) Vu-Topper RM

What does 'I' denotes in a geometric sequence?

➤ **Common ratio**

Question No:249 (Marks:1) Vu-Topper RM

Let $A=\{1,2,3,4\}$ and $B=\{7\}$ then the constant function from A to B is _____.

➤ **Both one to one and onto**

Question No:250 (Marks:1) Vu-Topper RM

$n!$ is defined to be _____.

➤ **The product of the integers from 1 to n**

Question No:251 (Marks:1) Vu-Topper RM

A set is called countable if, and only if, it is _____.

➤ **Both b and c**

Question No:252 (Marks:1) Vu-Topper RM

Which of the following is the example of an alternating sequence?

➤ **$C_n = n/n+1$ for $n \geq 0$**

Question No:253 (Marks:1) Vu-Topper RM

$x+a, x+3a, x+5a, \dots$ is an _____

➤ **Arithmetic sequence**

Question No:254 (Marks:1) Vu-Topper RM

$0, -5, -10, -15, \dots$ is an _____.

➤ **Arithmetic sequence**

Question No:255 (Marks:1) Vu-Topper RM

5, 9, 13, 17, ... is an _____.

➤ **Arithmetic sequence**

Question No:256 (Marks:1) Vu-Topper RM

An important data type in computer programming consists of _____.

➤ **Finite sequence**

Question No:257 (Marks:1) Vu-Topper RM

One-to-one correspondence means the condition of _____.

➤ **One-one and ontoboth**

Question No:258 (Marks:1) Vu-Topper RM

Cardinality means the total number of elements in a set.

➤ **True**

Question No:259 (Marks:1) Vu-Topper RM

Inverse of a function may not be a function.

➤ **True**

Question No:260 (Marks:1) Vu-Topper RM

The functions 'f' and 'g' are inverse of each other if and only if their composition gives _____.

➤ **Identity function**

Question No:261 (Marks:1) Vu-Topper RM

A set is called finite if , and only if, it is the _____ or there is _____.

➤ **Empty set or one-to-one**

Question No:262 (Marks:1) Vu-Topper RM

Let f and g be the two functions from R to R defined by $f(x) = |x|$ and $g(x) = \text{square root } x^2$ for all $x \in R$, then _____.

➤ **F(x) is not equal to g(x)**

Question No:263 (Marks:1) Vu-Topper RM

If $f^{-1}(x) = 6 - x/2$ then $f^{-1}(2)$ is _____.

➤ **2**

Question No:264 (Marks:1) Vu-Topper RM

Let $f(2)=3$, $g(2)=3$, $f(4)=1$ and $g(4)=2$ then the value of $f \circ g(4)$ is _____.

➤ **3**

Question No:265 (Marks:1) Vu-Topper RM

An important data type in computer programming consists of ____.

➤ **Finite sequence**

Question No:266 (Marks:1) Vu-Topper RM

Let $f(x)=x+3$ then $f^{-1}(x)$ is _____.

➤ **X-3**

Question No:267 (Marks:1) Vu-Topper RM

If $f:X \rightarrow Y$ and $g:Y \rightarrow Z$ are both onto function. Then $g \circ f : X \rightarrow Z$ is _____.

➤ **One-to-one function**

Question No:268 (Marks:1) Vu-Topper RM

The functions 'f' and 'g' are inverse of each other if and only if their composition gives

➤ **Identity function**

Question No:269 (Marks:1) Vu-Topper RM

The two function 'f' and 'g' are equal if _____.

➤ **$F(x) = 3x$ and $g(x) = 6x^2+3x/2x+1$ for all $x \in \mathbb{R}$**

Question No:270 (Marks:1) Vu-Topper RM

Two functions 'f' and 'g' from X To Y are said to be equal if and only if_____.

➤ **$F(x)$ and $g(x)$ for all 'x' belongs to X**

Question No:271 (Marks:1) Vu-Topper RM

Which of the following set is the domain of a sequence?

➤ **Set of natural numbers**

Question No:272 (Marks:1) Vu-Topper RM

If 1st term of a geometric sequence is 2 and common ratio is $1/2$, then 3rd term of sequence is

_____.

➤ **$1/2$**

Question No:273 (Marks:1) Vu-Topper RM

Composition of a function is a commutative operation.

➤ **True**

Question No:274 (Marks:1) Vu-Topper RM

If a function $(g \circ f)(x): X \rightarrow Z$ is defined as $(g \circ f)(x)=g(f(x))$ for all $x \in X$. Then the function _____ is known as composition of f and g.

➤ **$(g \circ f)$**

Question No:275 (Marks:1) Vu-Topper RM

A set is countably infinite if and only if and only if, it has the same cardinality as the set of

_____.

➤ **Positive integers**

Question No:276 (Marks:1) Vu-Topper RM

The 3rd term of the sequence $b_n=5^n$ is _____.

➤ **125**

Question No:277 (Marks:1) Vu-Topper RM

If f is a bijective function then $(f^{-1}(f(x)))$ is equal to _____.

➤ **X**

Question No:278 (Marks:1) Vu-Topper RM

Let $f(x) = x$ and $g(x) = -x$ for all $x \in \mathbb{R}$, then $(f+g)(x)$ is _____.

➤ **0**

Question No:279 (Marks:1) Vu-Topper RM

An infinite sequence may have only a finite number of values.

➤ **True**

Question No:280 (Marks:1) Vu-Topper RM

The functions $f \circ g$ and $g \circ f$ are always equal.

➤ **False**

Question No:281 (Marks:1) Vu-Topper RM

If f and g are two one-to-one functions, then their composition that is $g \circ f$ is one-to-one.

➤ **True**

Question No:282 (Marks:1) Vu-Topper RM

A function whose range consists of only one element is called _____.

➤ **Constant function**

Question No:283 (Marks:1) Vu-Topper RM

Let $X = \{1, 5, 9\}$ and $Y = \{3, 4, 7\}$. Define a function f from X to Y such that $f(1)=7$, $f(5)=3$, $f(9)=4$ then which of the following statement about ' f ' is true?

➤ **f is both one-to-one and onto**

Question No:284 (Marks:1) Vu-Topper RM

$Y = x^3$ is a graph of bijective function from $\mathbb{R}$ to $\mathbb{R}$.

➤ **True**

Question No:285 (Marks:1) Vu-Topper RM

A function whose inverse function exists is called a/an _____.

➤ **Invertible**

Question No:286 (Marks:1) Vu-Topper RM

Let F and g be two functions defined by $f(x) = x+2$ and $g(x) = 2x+1$. Then the composition of f and g is _____.

➤ **$2x + 5$**

Question No:287 (Marks:1) Vu-Topper RM

If r is a positive real number, then the value of r in $3, r, r, -27r$ is

➤ **-9**

Question No:288 (Marks:1) Vu-Topper RM

The _____ of the terms of a sequence forms a series.

➤ **Sum**

Question No:289 (Marks:1) Vu-Topper RM

The sum of first five whole number is _____.

➤ **10**

Question No:290 (Marks:1) Vu-Topper RM

If $f_k = f_{k-1} + f_{k-2}$ then $f_0 = 1$, $f_1 = 2$, then $f_2 =$ _____.

➤ **3**

Question No:291 (Marks:1) Vu-Topper RM

Let $f(x) = 3x$ and $g(x) = x + 2$ define functions f and g from R to R , then $(f.g)(x)$ is _____.

➤ **$3x^2 + 6x$**

Question No:292 (Marks:1) Vu-Topper RM

Let R be a relation on a set A . If R is reflexive then its compliment is _____.

➤ **Irreflexive**

Question No:293 (Marks:1) Vu-Topper RM

If $A =$ Set of students of virtual university then A has been written in the _____.

➤ **Descriptive form**

Question No:294 (Marks:1) Vu-Topper RM

If a function $(g \circ f)(x): X \rightarrow Z$ is defined as $(g \circ f)(x) = g(f(x))$ for all $x \in X$. Then the function _____ is known as composition of f and g .

➤ **$(g \circ f)$**

Question No:295 (Marks:1) Vu-Topper RM

If X and Y are independent random variables and a and b are constants, then $\text{Var}(aX + bY)$ is equal to

➤ **$a\text{Var}(X) + b\text{Var}(Y)$**

Question No:296 (Marks:1) Vu-Topper RM

Let $A = \{1, 2, 3\}$ and $B = \{2, 4\}$ then number of functions from A to B are _____.

➤ **8**

Question No:297 (Marks:1) Vu-Topper RM

p is equivalent to q means _____.

➤ **p is necessary and sufficient for q .**

Question No:298 (Marks:1) Vu-Topper RM

Let A and B be subsets of U with $n(A) = 12$, $n(B) = 15$, $n(A') = 17$, and $n(A \cap B) = 8$, then $n(U) =$ _____.

➤ **29**

Question No:299 (Marks:1) Vu-Topper RM

For the following relation to be a function, x can not be what values?

$$R = \{(2,4), (x,1), (4,2), (5,6)\}$$

➤x cannot be 2, 4 or 5

Question No:300 (Marks:1) Vu-Topper RM

Find the number of the word that can be formed of the letters of the word “ELEVEN”.

➤120